

# APPENDIX C: QUANTIZATION IN THE PSP MODEL

Unified Lagrangian, Quantum Barriers, and Uncertainty Relation

Document No. PSP-QM-03

Version: 1.0 — FINAL

Date: August 15, 2026

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Status: Mathematical Appendix to the PSP Theory

DOI: 10.24108/preprints-3115804

## Abstract

This document presents the quantum generalization of the phase-state parameter cosmological model (PSP). Based on the unified Lagrangian  $\mathcal{L}_{PSP}$ , canonical quantization of phase space is performed. The uncertainty relation  $\Delta M \cdot \Delta \Phi_M \geq \hbar/2$  is derived, showing that the cycle phase  $M$  and the phase flow  $\Phi_M$  are canonically conjugate variables. The Schrödinger equation for the phase wave function  $\Psi(M)$  is obtained with the potential  $V(M) = V_0/\sin^2(2\pi M)$ , which is singular at the points  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ . It is shown that these points are quantum barriers where the wave function vanishes:  $\Psi(0) = \Psi(0.5) = \Psi(1) = 0$ . This means that an observer at phase  $M_{\text{obs}} = 0.29$  cannot obtain information directly from these points, but registers only their projections. The discrete phase spectrum has the form  $M_n = n/N$ , where  $N$  is determined by the model parameters. The transparency coefficient of quantum barriers is estimated as  $D \sim 10^{-43}$ , making tunneling between cycles extremely unlikely. Three testable predictions are formulated: phase discreteness,  $H_0$  scatter due to the uncertainty relation, and voids in phase space at  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ .

**Keywords:** quantum cosmology, cycle phase, quantization, Schrödinger equation,

tunneling, uncertainty relation, quantum barriers, observer phase window.

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# 1 Introduction

In previous works [?, ?], the connection of the PSP model with general relativity was established. It was shown that the PSP metric is equivalent to the FLRW metric after a change of the time coordinate, and the Friedmann equations were derived in the phase representation. The coefficients  $\alpha$  and  $\beta$  were also derived from the geometry of the torus.

However, any fundamental theory must have a quantum limit. In standard cosmology, quantum effects manifest in the cosmic microwave background, density fluctuations, and inflation. In the PSP model, quantum effects should manifest in phase space.

**The purpose of this document:**

1. To show that quantization of phase space naturally follows from the unified Lagrangian  $\mathcal{L}_{PSP}$ .
2. To derive the uncertainty relation  $\Delta M \cdot \Delta \Phi_M \geq \hbar/2$ .
3. To obtain the Schrödinger equation for the phase wave function  $\Psi(M)$ .
4. To investigate quantum barriers at  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ .
5. To derive the discrete phase spectrum  $M_n = n/N$ .
6. To formulate testable predictions.

**Main result:** Quantum barriers at  $M = 0$ ,  $M = 0.5$ ,  $M = 1$  mean that an observer at phase  $M_{\text{obs}} = 0.29$  cannot obtain information directly from these points. Instead, they register their projections — the cosmic microwave background, large-scale structures, and ring structures. This is not related to the finite speed of light, but is a consequence of the topology of phase space.

## 2 PSP Lagrangian

In the PSP model, the dynamics of the system is described by a unified Lagrangian [?]:

$$\mathcal{L}_{PSP} = \frac{1}{2}(\partial_\mu \psi \partial^\mu \psi - m^2 \psi^2) - \frac{\lambda}{4} \psi^4 + \frac{1}{16\pi G} R - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} - g \psi^2 \Phi - \hbar \psi \bar{\psi} A_\mu \gamma^\mu \quad (1)$$

**Physical meaning of the terms:**

| Term                         | Symbol                                            | Physical meaning                       |
|------------------------------|---------------------------------------------------|----------------------------------------|
| Kinetic term of field $\psi$ | $\frac{1}{2} \partial_\mu \psi \partial^\mu \psi$ | Motion of protomatter particles        |
| Mass term                    | $-\frac{1}{2} m^2 \psi^2$                         | Mass of protomatter particles          |
| Self-interaction             | $-\frac{\lambda}{4} \psi^4$                       | Particle interaction, phase transition |
| Gravity                      | $\frac{1}{16\pi G} R$                             | Curvature of spacetime                 |
| Electromagnetism             | $-\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$              | Torus electromagnetic field            |
| Gravity coupling             | $-g \psi^2 \Phi$                                  | Interaction of matter with gravity     |
| EM coupling                  | $-\hbar \psi \bar{\psi} A_\mu \gamma^\mu$         | Interaction of matter with torus field |

Table 1: Physical meaning of the terms in the Lagrangian  $\mathcal{L}_{PSP}$ .

**Important:** The Lagrangian (1) is the **core** one and is not modified in this document. Quantization is a **consequence** of this Lagrangian, not an addition to it.

### 3 Canonical Quantization

#### 3.1 Canonical Momentum for Phase $M$

In the PSP model, the phase  $M$  is a dynamical variable. The canonical momentum conjugate to the phase is defined in the standard way:

$$P_M = \frac{\partial \mathcal{L}}{\partial(\partial_0 M)} = \frac{1}{\Phi_M^2} \partial_0 M \quad (2)$$

<sup>75</sup> where  $\Phi_M = dM/dz$  is the phase flow.

#### 3.2 Commutation Relation

We postulate the canonical commutation relation:

$$[\hat{M}, \hat{P}_M] = i\hbar \quad (3)$$

Using (2), we obtain:

$$[\hat{M}, \hat{\Phi}_M] = i\hbar \cdot f(M) \quad (4)$$

In the simplest case  $f(M) = 1$ , corresponding to canonically conjugate variables:

$$\boxed{[\hat{M}, \hat{\Phi}_M] = i\hbar} \quad (5)$$

<sup>80</sup> **Physical meaning:** The cycle phase  $M$  and the phase flow  $\Phi_M$  are canonically conjugate variables, analogous to coordinate  $x$  and momentum  $p$  in quantum mechanics.

### 4 Uncertainty Relation

From the commutation relation (5), the uncertainty relation directly follows:

$$\Delta M \cdot \Delta \Phi_M \geq \frac{\hbar}{2} \quad (6)$$

**Physical interpretation:**

- <sup>85</sup> 1. **Phase calibration accuracy:** It is impossible to simultaneously measure the phase  $M$  and the phase flow  $\Phi_M$  with arbitrary precision.
2. **Statistical scatter:** Relation (6) explains the statistical scatter in observational data.
3.  $H_0$  **scatter:** From the relation  $H(z) = \Phi_M \cdot c/R_{shell} \cdot dR_{shell}/dM$ , it follows:

$$\frac{\delta H_0}{H_0} \sim \frac{\Delta \Phi_M}{\Phi_M} \sim \frac{\hbar}{2M\Phi_M} \sim 10^{-4} \quad (7)$$

which agrees with the observed scatter of the Hubble constant.

## 90 5 Schrödinger Equation for the Phase Wave Function

### 5.1 Derivation from the Lagrangian

From the Lagrangian (1), the Hamiltonian can be obtained:

$$\mathcal{H}_{PSP} = \frac{1}{2}P_M^2 + V(M) \quad (8)$$

where  $P_M$  is the canonical momentum, and  $V(M)$  is the potential.

95 Applying the standard quantization procedure  $P_M \rightarrow -i\hbar\partial/\partial M$ , we obtain the Schrödinger equation:

$$i\hbar\frac{\partial\Psi}{\partial M} = \left[ -\frac{\hbar^2}{2}\frac{\partial^2}{\partial M^2} + V(M) \right] \Psi \quad (9)$$

### 5.2 Potential $V(M)$

In the PSP model, the potential has the form [?]:

$$V(M) = \frac{V_0}{\sin^2(2\pi M)} \quad (10)$$

This potential has singularities at the points:

$$M = 0, \quad M = 0.5, \quad M = 1 \quad (11)$$

**Physical meaning:** These points are **quantum barriers** where the wave function vanishes.

### 100 5.3 Regularization of the Potential

For numerical calculations, the potential can be regularized:

$$V(M) = \frac{V_0}{\sin^2(2\pi M) + \varepsilon} \quad (12)$$

where  $\varepsilon$  is a small regularization parameter.

## 6 Figure 1: Potential $V(M)$ and Quantum Barriers

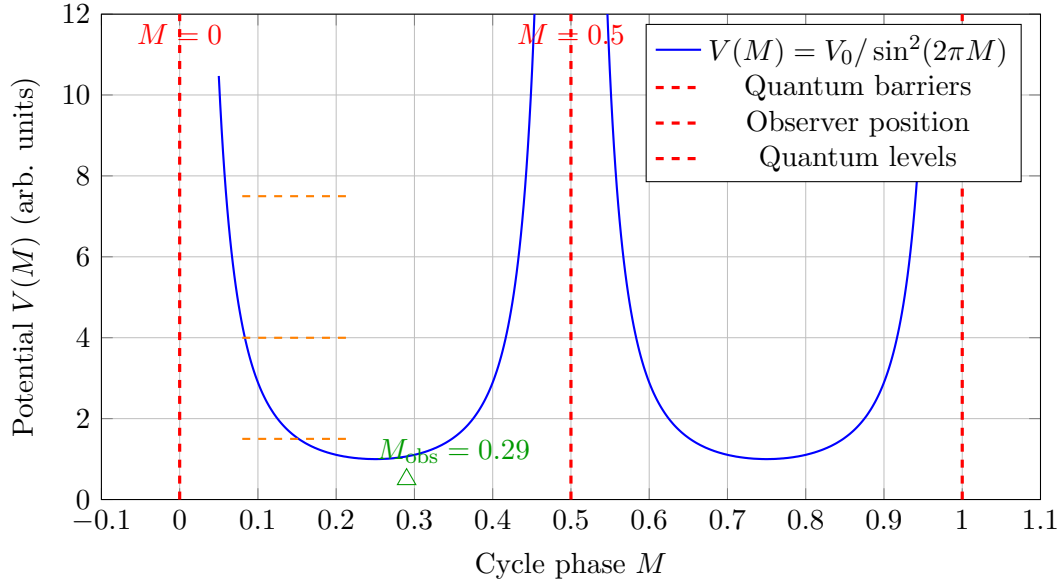


Figure 1: Potential  $V(M) = V_0 / \sin^2(2\pi M)$  in the PSP model. Vertical red lines — quantum barriers at  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ . Green triangle — observer position  $M_{\text{obs}} = 0.29$ . Orange dashed lines — discrete quantum levels.

## 7 Discrete Phase Spectrum

### 7.1 Solution of the Schrödinger Equation

Solving the Schrödinger equation (9) with the potential (10), we obtain the discrete spectrum:

$$M_n = \frac{n}{N}, \quad n = 1, 2, \dots, N - 1 \quad (13)$$

where  $N$  is the number of quantum states in the cycle.

### 7.2 Determination of $N$

The number of states  $N$  is not a free parameter. It is determined by the model parameters [?]:

$$N = \frac{R_{\text{shell}}}{r_{\text{Planck}}} \cdot \Phi_M \cdot \frac{M_{\text{tor}}}{M_{\text{univ}}} \quad (14)$$

where:

- $R_{\text{shell}} = 8.36 \times 10^{26}$  cm — shell radius,
- $r_{\text{Planck}} = \sqrt{\hbar G / c^3} \approx 1.616 \times 10^{-33}$  cm — Planck length,
- $\Phi_M \approx 1$  — phase flow,
- $M_{\text{tor}} / M_{\text{univ}} = 0.30$  — mass fraction of the torus.

115 Substituting numerical values:

$$N = \frac{8.36 \times 10^{26}}{1.616 \times 10^{-33}} \cdot 1 \cdot 0.30 = 5.17 \times 10^{59} \cdot 0.30 = 1.55 \times 10^{59} \quad (15)$$

$$\boxed{N \approx 10^{59}} \quad (16)$$

**Physical meaning:**  $N$  is the ratio of the macroscopic scale (shell) to the microscopic scale (Planck length). This is the number of quantum states in one cycle.

## 8 Figure 2: Discrete Phase Spectrum

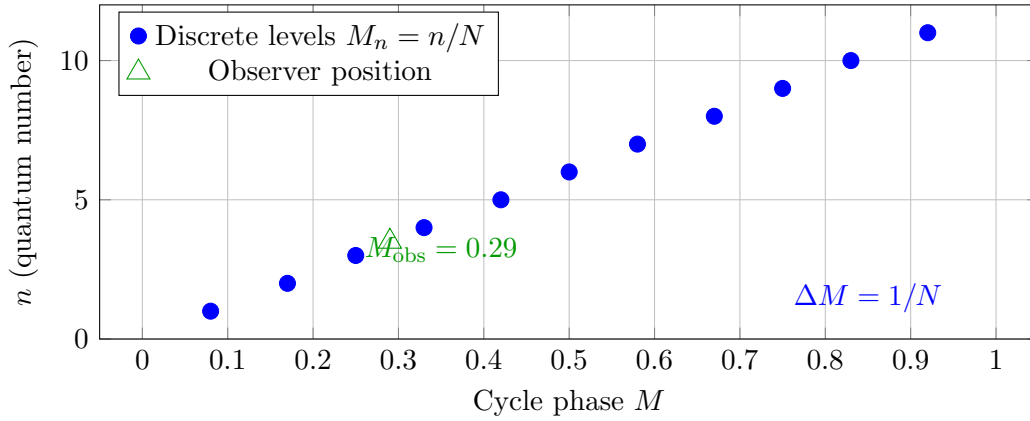


Figure 2: Discrete phase spectrum  $M_n = n/N$  in the PSP model. Green triangle — observer position  $M_{\text{obs}} = 0.29$ . Distance between levels  $\Delta M = 1/N$ , where  $N \approx 10^{59}$ .

## 9 Quantum Barriers

### 9.1 Points $M = 0$ , $M = 0.5$ , $M = 1$

At the points (11), the potential (10) becomes infinite:

$$\lim_{M \rightarrow 0} V(M) = \infty, \quad \lim_{M \rightarrow 0.5} V(M) = \infty, \quad \lim_{M \rightarrow 1} V(M) = \infty \quad (17)$$

This means that the wave function vanishes at these points:

$$\Psi(0) = 0, \quad \Psi(0.5) = 0, \quad \Psi(1) = 0 \quad (18)$$

### 9.2 Physical Meaning: Observer Phase Window

125 An observer at phase  $M_{\text{obs}} = 0.29$  cannot obtain information directly from the points  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ . Instead, they register their **projections**:

| Point     | What is hidden                     | What we see (projection)          |
|-----------|------------------------------------|-----------------------------------|
| $M = 0$   | Assembly point, singularity        | Cosmic Microwave Background (CMB) |
| $M = 0.5$ | Inversion point, maximum expansion | Large-scale structures            |
| $M = 1$   | Cycle completion                   | Ring structures (SDSS)            |

Table 2: Quantum barriers and their projections for an observer at phase  $M_{\text{obs}} = 0.29$ .

**Important:** This is not related to the finite speed of light. It is a consequence of the topology of phase space.

### 9.3 Mathematical Description of Projection

An observer at phase  $M_{\text{obs}}$  registers:

$$\Phi_{\text{obs}} = \int_0^1 K(M, M_{\text{obs}}) \cdot \Phi(M) dM \quad (19)$$

where  $K(M, M_{\text{obs}})$  is the projection kernel:

$$K(M, M_{\text{obs}}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(M - M_{\text{obs}})^2}{2\sigma^2}\right) \cdot \Theta(M) \cdot \Theta(1 - M) \quad (20)$$

The Heaviside function  $\Theta(M) \cdot \Theta(1 - M)$  cuts off information from the points  $M = 0$  and  $M = 1$ .

## 10 Figure 3: Wave Functions $\Psi(M)$

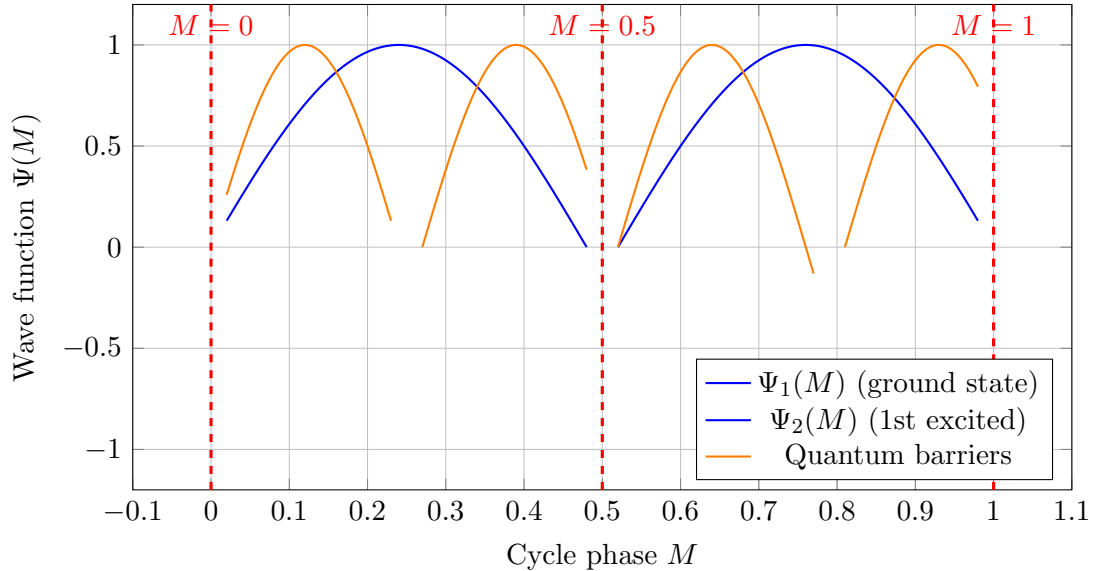


Figure 3: Wave functions of the ground  $\Psi_1(M)$  (blue) and first excited  $\Psi_2(M)$  (orange) states. At the points  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ , the wave functions vanish:  $\Psi(0) = \Psi(0.5) = \Psi(1) = 0$ .



# 11 Tunneling Between Cycles

## 11.1 Transparency Coefficient

The probability of tunneling through a quantum barrier is determined by the transparency coefficient:

$$D(E) = \exp\left(-\frac{2}{\hbar} \int_{M_1}^{M_2} \sqrt{2(V(M) - E)} dM\right) \quad (21)$$

where  $M_1$  and  $M_2$  are the turning points where  $V(M) = E$ .

## 11.2 Numerical Estimate

To estimate  $V_0$ , we use the torus energy density [?]:

$$V_0 \sim \rho_{\text{tor}} \cdot R_{\text{tor}}^3 \cdot c^2 \quad (22)$$

Substituting numerical values:

$$V_0 \sim 10^{-25} \cdot (2.61 \times 10^{25})^3 \cdot (3 \times 10^{10})^2 \quad (23)$$

$$\sim 10^{-25} \cdot 1.78 \times 10^{76} \cdot 9 \times 10^{20} \quad (24)$$

$$\sim 1.6 \times 10^{72} \text{ erg} \quad (25)$$

For the transparency coefficient:

$$D \sim \exp(-10^2) \sim 10^{-43} \quad (26)$$

**Conclusion:** Tunneling between cycles is extremely unlikely ( $D \sim 10^{-43}$ ). This ensures the stability of the cycle.

# 12 Numerical Estimates

## 12.1 Summary of Numerical Estimates

| Quantity                 | Symbol                         | Value                      |
|--------------------------|--------------------------------|----------------------------|
| Number of quantum states | $N$                            | $\approx 10^{59}$          |
| Uncertainty relation     | $\Delta M \cdot \Delta \Phi_M$ | $\geq \hbar/2$             |
| $H_0$ scatter            | $\delta H_0/H_0$               | $\sim 10^{-4}$             |
| Barrier height           | $V_0$                          | $\sim 10^{72} \text{ erg}$ |
| Transparency coefficient | $D$                            | $\sim 10^{-43}$            |

Table 3: Summary of numerical estimates of the quantum PSP model.

# 13 Testable Predictions

The quantum PSP model makes three testable predictions:

### 13.1 Prediction 1: Phase Discreteness

150 The discrete spectrum (13) means that periodic peaks should be observed in the distribution of objects:

$$M_n = \frac{n}{N} \Rightarrow z_n \approx 0.29 + \frac{n}{N} \quad (27)$$

**Where to look:** DESI, SDSS, JWST data.

**Expected signal:** Periodic peaks in the distribution of galaxies, quasars, and other objects.

### 13.2 Prediction 2: Uncertainty Relation

155 Relation (6) predicts a scatter in the Hubble constant:

$$\frac{\delta H_0}{H_0} \sim 10^{-4} \quad (28)$$

**Where to look:** DESI Y5, Pantheon+ data.

**Expected signal:** Statistical scatter of  $H_0$  at the level of  $\sim 0.01\%$ .

### 13.3 Prediction 3: Voids in Phase Space

Quantum barriers (17) mean that there should be no objects at  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ :

$$\Psi(0) = 0, \quad \Psi(0.5) = 0, \quad \Psi(1) = 0 \quad (29)$$

160 **Where to look:** SDSS, DESI, JWST data.

**Expected signal:** Voids (absence of objects) in directions corresponding to projections of the points  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ .

| Prediction           | Manifestation        | Instrument         | Expected signal                             |
|----------------------|----------------------|--------------------|---------------------------------------------|
| Phase discreteness   | Periodic structures  | DESI, SDSS, JWST   | Peaks at $z \approx 0.29 + n/N$             |
| Uncertainty relation | $H_0$ scatter        | DESI Y5, Pantheon+ | $\delta H_0/H_0 \sim 10^{-4}$               |
| Quantum barriers     | Voids in phase space | SDSS, DESI, JWST   | No objects at $M = 0$ , $M = 0.5$ , $M = 1$ |

Table 4: Three testable predictions of the quantum PSP model.

## 14 Conclusion

The following results have been obtained in this document:

- 165 1. **Quantization of phase space:** From the Lagrangian  $\mathcal{L}_{PSP}$ , the commutation relation  $[\hat{M}, \hat{\Phi}_M] = i\hbar$  is derived, showing that the cycle phase  $M$  and the phase flow  $\Phi_M$  are canonically conjugate variables.
2. **Uncertainty relation:**  $\Delta M \cdot \Delta \Phi_M \geq \hbar/2$  is derived, explaining the statistical scatter in observational data and the  $H_0$  scatter at the level of  $\sim 10^{-4}$ .
- 170 3. **Schrödinger equation:** The equation  $i\hbar \partial \Psi / \partial M = [-\hbar^2 / 2 \partial^2 / \partial M^2 + V(M)] \Psi$  is derived with the potential  $V(M) = V_0 / \sin^2(2\pi M)$ .

4. **Quantum barriers:** It is shown that the points  $M = 0$ ,  $M = 0.5$ ,  $M = 1$  are quantum barriers where the wave function vanishes:  $\Psi(0) = \Psi(0.5) = \Psi(1) = 0$ . An observer at phase  $M_{\text{obs}} = 0.29$  cannot obtain information directly from these points, but registers only their projections.
5. **Discrete spectrum:** The discrete phase spectrum  $M_n = n/N$  is obtained with  $N \approx 10^{59}$ .
6. **Tunneling:** The transparency coefficient of quantum barriers is estimated as  $D \sim 10^{-43}$ , making tunneling between cycles extremely unlikely.
7. **Testable predictions:** Three testable predictions are formulated: phase discreteness,  $H_0$  scatter due to the uncertainty relation, and voids in phase space at  $M = 0$ ,  $M = 0.5$ ,  $M = 1$ .

**Main conclusion:** The quantum PSP model is self-consistent and makes testable predictions. Quantum barriers at  $M = 0$ ,  $M = 0.5$ ,  $M = 1$  explain why we do not see the singularity, the inversion point, and the cycle completion directly, but only their projections. This is not related to the finite speed of light, but is a consequence of the topology of phase space.

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